HDOC Day-51

Question

For each array element, find the nearest greater element on its right. Print -1 if none exists. Display results separated by commas, using pointers and nested loops.

C Program

```
#include <stdio.h>

void next_greater(int *arr, int n)
{
    int i, j, ans;
    for (i = 0; i < n; i++) {
        ans = -1;
        for (j = i + 1; j < n; j++) {
            if (*(arr + j) > *(arr + i)) {
                ans = *(arr + j);
                break;
            }
        }
        printf("%d", ans);
        if (i < n - 1) printf(", ");
    }
    printf("\n");
}

int main()
{
    int n, i;
    scanf("%d", &n);
    if (n <= 0) return 0;
    int arr[n];
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    next_greater(arr, n);
    return 0;
}
```

Sample Input

```
5
2 5 3 4 1
```

Sample Output

```
5, 5, -1, 4, -1
```

Explanation

The outer loop selects each element. The inner loop searches from the closest neighboring position outward and stops at the first greater value. If none is found, the answer remains -1. The array is passed to the function through a pointer.